Experiment Number : 07 **Date:**

METHOD OVERLOADING AND METHOD OVERRIDING

Aim:

To understand and implement method overloading and method overriding.

PRE LAB EXERCISE

QUESTIONS

- ✓ **What is method overloading?**

  Defining multiple methods with the same name but different parameters in the same class.

- ✓ **What is method overriding?**

  Redefining a parent class method in a child class with the same signature.

- ✓ **Difference between overloading and overriding.**

  Overloading is compile-time polymorphism with different parameters, while overriding is runtime polymorphism with the same method signature in inheritance.

IN LAB EXERCISE

Objective:

To demonstrate compile-time and runtime polymorphism.

PROGRAMS:

1.Student Result System (Method Overriding)

Description:

- Base class Student has method display Result().
- Subclasses UG Student and PG Student override the method to show different grading systems.

Code :

```
import java.util.Scanner;
```

```
// Base class
class Student {
    String name;

    void displayResult() {
        System.out.println("Student Result");
    }
}

// UG Student subclass
class UGStudent extends Student {
    int marks;

    UGStudent(String n, int m) {
        name = n;
        marks = m;
    }

    @Override
    void displayResult() {
        double percentage = (marks / 100.0) * 100;
        System.out.println("UG Student: " + name);
        System.out.println("Marks: " + marks);
        System.out.println("Percentage: " + percentage + "%");
    }
}

// PG Student subclass
class PGStudent extends Student {
    double gpa;
```

```
    PGStudent(String n, double g) {
        name = n;
        gpa = g;
    }

    @Override
    void displayResult() {
        System.out.println("PG Student: " + name);
        System.out.println("GPA: " + gpa + " / 10");
    }
}

// Main class
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        // Input for UG student
        System.out.print("Enter UG Student Name: ");
        String ugName = sc.nextLine();
        System.out.print("Enter UG Student Marks (out of 100): ");
        int ugMarks = sc.nextInt();
        sc.nextLine(); // consume newline

        // Input for PG student
        System.out.print("Enter PG Student Name: ");
        String pgName = sc.nextLine();
        System.out.print("Enter PG Student GPA (0-10): ");
        double pgGpa = sc.nextDouble();

        // Create objects
```

```
        Student s1 = new UGStudent(ugName, ugMarks);
        Student s2 = new PGStudent(pgName, pgGpa);

        System.out.println("\n--- Student Results ---");
        s1.displayResult();
        System.out.println();
        s2.displayResult();

        sc.close();
    }
}
```

OUTPUT:

Sample Input:

Enter UG Student Name: Ram

Enter UG Student Marks (out of 100): 85

Enter PG Student Name: Ravi

Enter PG Student GPA (0-10): 9.2

Output:

--- Student Results ---

UG Student: Ram

Marks: 85

Percentage: 85.0%

PG Student: Ravi

GPA: 9.2 / 10

```
Enter UG Student Name: Santhosh
Enter UG Student Marks (out of 100): 97
Enter PG Student Name: TMA
Enter PG Student GPA (0-10):
7

--- Student Results ---
UG Student: Santhosh
Marks: 97
Percentage: 97.0%

PG Student: TMA
GPA: 7.0 / 10
```

2. Calculator Program (Method Overloading)

Description:
Create a Calculator class with multiple add() methods to calculate:

- Addition of 2 integers
- Addition of 3 integers
- Addition of 2 double numbers

Code:

```
import java.util.Scanner;
class Calculator {
    int add(int a, int b) {
        return a + b;
    }

    int add(int a, int b, int c) {
        return a + b + c;
    }

    double add(double a, double b) {
        return a + b;
    }
}
```

```
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        Calculator calc = new Calculator();

        System.out.print("Enter two integers: ");
        int x = sc.nextInt();
        int y = sc.nextInt();
        System.out.println("Sum of two integers: " + calc.add(x, y));

        System.out.print("Enter three integers: ");
        int p = sc.nextInt();
        int q = sc.nextInt();
        int r = sc.nextInt();
        System.out.println("Sum of three integers: " + calc.add(p, q, r));

        System.out.print("Enter two decimal numbers: ");
        double a = sc.nextDouble();
        double b = sc.nextDouble();
        System.out.println("Sum of two doubles: " + calc.add(a, b));

        sc.close();
    }
}
```

Output:

```
Enter two integers: 5 7
Sum of two integers: 12
Enter three integers: 1 4 5
Sum of three integers: 10
Enter two decimal numbers: 9.3 2.5
Sum of two doubles: 11.8
```

Sample Input:

Enter two integers: 10 20

Enter three integers: 5 10 15

Enter two decimal numbers: 2.5 3.5

Output:

Sum of two integers: 30

Sum of three integers: 30

Sum of two doubles: 6.0

POST LAB EXERCISE

- ✓ **Is return type important in method overloading and method overriding?**
  Yes, return type is not important in overloading but must be same or covariant in overriding
- ✓ **Can you overload a method by changing only the return type?**
  No, a method cannot be overloaded by changing only the return type.
- ✓ **Can static methods be overridden? Can they be overloaded?**
  Static methods cannot be overridden but can be overloaded.
- ✓ **Can a method be overridden if the parameter list is different?**

  No, a method cannot be overridden if the parameter list is different.

Result:

Thus the method overloading and overriding concepts were implemented and executed successfully.

ASSESSMENT

Description	Max Marks	Marks Awarded
Pre Lab Exercise	**5**	
In Lab Exercise	**10**	
Post Lab Exercise	**5**	
Viva	**10**	
Total	**30**	
Faculty Signature		